

The Complexity–Spuriousness Trade-off in Probing: Toward a Unified Capacity Measure for Diagnostic and Causal Interpretability

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Abstract

Two recent results paint an uncomfortable picture of how we interpret neural networks. On the diagnostic side, [Pimentel et al. \(2020\)](#) show that, under an information-theoretic operationalisation, a sufficiently powerful probe will *always* recover a target property—making the success of probing uninformative about the representation itself. On the causal side, [Sutter et al. \(2025\)](#) show that, with a sufficiently complex alignment map, *any* algorithm can be cast as a distributed abstraction of *any* network—making causal abstraction vacuous without constraints. Both results are *population-level* statements about a limit of infinite probe capacity. In practice, however, we never operate in that limit: we fit a probe or an alignment map on a finite sample, and the danger is the *finite-sample optimism* that this fitting introduces—a propensity to find structure that is not there. In this technical report we argue that this propensity is, for both techniques, an instance of the same object: the **Rademacher complexity** of the interpretability method’s hypothesis class, evaluated on the representations. We make this unification precise, separate it cleanly from the population-level vacuity that the two theorems describe, and propose a concrete experimental protocol—building directly on an existing implementation of the hierarchical equality task—to measure how this *spurious-fitting capacity* scales with probe complexity for diagnostic probing and for distributed alignment search (DAS). Our central question is not *whether* both methods fail in the limit (they do), but *how gracefully* each approaches that limit, and which structural constraints govern the rate.

1 Motivation

A probe is a supervised model trained to predict some property T from a network’s representations R ; if the probe succeeds, we are tempted to conclude that the representations *encode* the property ([Alain and Bengio, 2017](#); [Hewitt and Manning, 2019](#); [Belinkov, 2022](#)). [Pimentel et al. \(2020\)](#) recast this practice information-theoretically: probing estimates the mutual information $I(T; R)$, and the best estimate is obtained by the *best-performing* probe one can train, regardless of its complexity. The corollary is uncomfortable: under their [section 2](#) assumptions, a contextual representation contains exactly as much information about a word-level property as the underlying sentence does, so a powerful enough probe recovers the property for reasons that need have nothing to do with how—or whether—the network uses it.

Causal probing was meant to escape this critique by intervening rather than merely decoding. Distributed alignment search (DAS; [Geiger et al., 2024](#)) learns a (possibly non-linear) alignment map ϕ onto the hidden space, performs interchange interventions in the aligned basis, and measures

the *interchange-intervention accuracy* (IIA) with which the network reproduces a target algorithm’s counterfactual behaviour. Sutter et al. (2025) show that this escape is illusory: once the linearity constraint on ϕ is lifted, *any* algorithm can be aligned to *any* network with near-perfect IIA—even randomly initialised networks that cannot solve the task. They call this the **non-linear representation dilemma**: lifting the linearity constraint removes any principled way to trade off the alignment map’s complexity against its accuracy.

Read together, the two results describe *the same failure with two faces*. In both, an interpretability method reports a score—probe accuracy, or IIA—that is optimised over a family of extraction functions, and in both, that score becomes uninformative once the family is expressive enough. The practitioner is left without a principled way to choose the family’s complexity. This report asks whether the two methods are not just analogous but *measurable on a common axis*: can we define a single notion of capacity, in the spirit of Rademacher complexity (Bartlett and Mendelson, 2002) or VC dimension (Vapnik, 1998), that quantifies each method’s propensity to find spurious structure, and study how that propensity grows?

2 Setup and notation

We adopt notation compatible with the two companion papers. A network induces a representation map $x \mapsto R$, with $R \in \mathbb{R}^d$. An **interpretability method** $\mathcal{M}$ is characterised by three ingredients: (i) a *hypothesis class* $\mathcal{H}_c$ indexed by a scalar *complexity knob* $c \in \mathbb{R}_+$; (ii) a per-example *fit* functional $\text{fit}(h; \cdot) \in [0, 1]$ measuring how well a hypothesis $h \in \mathcal{H}_c$ reproduces a target on one example; and (iii) a *selection rule* that, given a finite sample $\mathcal{D}_n$ of size n , returns $\hat{h} = \arg \max_{h \in \mathcal{H}_c} \hat{\text{fit}}(h; \mathcal{D}_n)$ and reports the achieved fit $s(\mathcal{M}) = \hat{\text{fit}}(\hat{h}; \mathcal{D}_n)$.

The two methods instantiate this template as follows.

Diagnostic probing. $\mathcal{H}_c$ is a family of probes $q_{\theta}(t \mid \mathbf{r})$ —e.g. multi-layer perceptrons whose depth or width grows with c , as in Pimentel et al. (2020). The targets are property labels t , the inputs are activations $\mathbf{r}$, and $\hat{\text{fit}}$ is accuracy (or, in the information-theoretic view, the negative cross-entropy that lower-bounds $I(T; R)$). The dataset is $\mathcal{D}_n = \{(\mathbf{r}_i, t_i)\}_{i=1}^n$.

Causal probing (DAS). $\mathcal{H}_c$ is a family of bijective alignment maps ϕ —e.g. reversible networks (Gomez et al., 2017) whose depth L_{rn} or hidden size d_{rn} grows with c , as in Sutter et al. (2025). The “label” of an example is the counterfactual output the *target algorithm* prescribes, the inputs are (base, source) pairs, and $\hat{\text{fit}}$ is the IIA: the rate at which the *frozen* network, intervened upon in the aligned basis, reproduces that output. The dataset is $\mathcal{D}_n = \{(\mathbf{x}_i^{\text{base}}, \mathbf{x}_i^{\text{src}}, \mathbf{y}_i^{\text{cf}})\}_{i=1}^n$.

The crucial structural difference—which we will argue is the heart of the matter—is that for DAS the hypothesis is not a classifier of $\mathbf{r}$. It is a *reparametrisation* ϕ of the hidden space, after which a *fixed* intervention and the *fixed* downstream network produce the prediction. The induced labelling function is $x \mapsto N(\text{intervene}_{\phi}(x))$, and the family is constrained to invertible ϕ .

3 A unified capacity measure

3.1 Spuriousness as fit to randomised targets

The propensity to “find what you want” is, statistically, the gap between the fit a method reports and the fit it *deserves*: the optimism of an empirical-risk maximiser. Uniform-convergence theory names the leading term of this optimism the *Rademacher complexity* of the hypothesis class (Bartlett and

Mendelson, 2002). We take this as our definition of spurious-fitting capacity, expressed directly in the accuracy space the two methods report.

Definition 1 (Spurious-fitting capacity). *Fix a representation sample of size n and let σ denote a target randomisation: a relabelling of the examples that destroys any genuine relationship between inputs and targets while preserving their marginals. The spurious-fitting capacity of a method $\mathcal{M}$ at complexity c is*

$$\mathfrak{R}_n(\mathcal{H}_c) := \mathbb{E}_{\sigma} \left[\sup_{h \in \mathcal{H}_c} \widehat{\text{fit}}(h; \sigma(\mathcal{D}_n)) \right] - s_{\text{chance}}, \quad (1)$$

i.e. the expected best fit the class achieves on randomised targets, centred at chance performance.

Equation 1 is, up to a scaling constant and the usual move from a margin loss to 0/1 accuracy, the empirical Rademacher complexity of $\mathcal{H}_c$ on the representation sample. Its appeal is threefold. First, it is *method-agnostic*: it specialises to diagnostic probing (random token labels) and to DAS (a random target algorithm, or equivalently permuted counterfactual outputs) without changing form. Second, it is exactly the quantity practitioners already estimate: it is the control-task accuracy of Hewitt and Liang (2019), and the random-network alignment of Sutter et al. (2025) is its causal analogue.¹ Third, it is *cheap*: a Monte-Carlo estimate requires only re-running the existing fitting pipeline on permuted targets.

3.2 Two failure modes, kept separate

It is essential not to conflate 1 with the failure the two theorems describe. They live at different scales.

- (a) **Population vacuity.** As $c \rightarrow \infty$, the *population* estimand becomes uninformative: an arbitrarily powerful probe saturates the mutual-information bound (Pimentel et al., 2020, Thm. 1), and an arbitrarily complex alignment map achieves perfect IIA on any network (Sutter et al., 2025, Thm. 1). In this limit *both methods are equally doomed*; the limit itself does not distinguish them.
- (b) **Finite-sample optimism.** At finite n , the reported score exceeds the population value by an amount controlled by $\mathfrak{R}_n(\mathcal{H}_c)$. This term *does* distinguish the methods, and it is the only thing that can.

We therefore argue that the scientifically productive question is not the $c \rightarrow \infty$ limit—which the theorems have settled—but the *rate* at which $\mathfrak{R}_n(\mathcal{H}_c)$ grows as we approach it. “Failing gracefully” is a statement about this rate. This separation also disarms an obvious objection (“your own theorems say the limit is degenerate, so why measure capacity?”): we are not studying the limit, we are studying the finite-sample road to it.

3.3 What “graceful” should mean, and the comparability problem

A natural temptation is to compare the two methods by plotting their capacity against a shared notion of “size”—say, parameter count—and asking whether one grows like $\mathcal{O}(\sqrt{P})$ where the other grows like $\mathcal{O}(P)$. We caution against taking the exponent itself as the contribution, for two reasons. First, generic norm- or parameter-based bounds yield the *same* $\mathcal{O}(\sqrt{P/n})$ form for any reasonable

¹We deliberately phrase 1 in accuracy space rather than as the textbook $\mathbb{E}_{\sigma}[\sup_h \frac{1}{n} \sum_i \sigma_i h(\mathbf{r}_i)]$ so that it is directly comparable across the two methods and directly estimable from the existing code paths, which already train on randomised labels.

class (Neyshabur et al., 2015), so an a priori difference in exponent is not expected; a measured difference would have to come from structure, not from counting. Second, “depth of an MLP probe” and “depth of a RevNet alignment map” are not the same unit, so a raw c -axis is not commensurable.

Both problems are solved by using *capacity itself* as the common axis. Concretely, we propose to plot, for each complexity level, the **signal** (fit on *correct* targets) against the **spurious capacity** (fit on *randomised* targets, 1). Each method traces a curve in this *signal–spurious plane* as c varies. A method **fails gracefully** if its curve hugs the top-left—high signal recovered per unit of spurious capacity spent—and **fails sharply** if signal and spurious capacity rise together. This operationalises “one method fails more gracefully than the other” without ever committing to incommensurable units, and it directly generalises the selectivity of Hewitt and Liang (2019) (which is the single-point gap signal – spurious) into a *scaling curve*.

4 A mechanism for divergence

If the two methods do diverge in the signal–spurious plane, the explanation must lie in structural constraints that one method imposes and the other does not. We identify three constraints that are present in DAS but absent in diagnostic probing, each of which should *throttle* spurious capacity below what the raw parameter count of ϕ would suggest:

1. **Invertibility.** The alignment map ϕ is constrained to be a bijection. A bijection cannot collapse or discard directions of the hidden space to manufacture a fit; it can only rotate and warp them. This is a genuine capacity restriction with no analogue in an unconstrained probe.
2. **The frozen network.** The prediction is produced by the *fixed* downstream network applied to an intervened, ϕ -aligned hidden state. The map cannot directly emit a label; it can only reroute information that the frozen weights will then process. Spurious fits must survive this fixed computation.
3. **Multi-intervention consistency.** A single ϕ must simultaneously satisfy the counterfactual constraints of *several* intervention types (in the hierarchical equality task, the two sub-equalities and their conjunction). These are many coupled constraints, not one label, and jointly satisfying them on randomised targets is harder than fitting a single randomised label.

Our working hypothesis is that these constraints make $\mathfrak{R}_n^{\text{DAS}}(c)$ grow *more slowly* than $\mathfrak{R}_n^{\text{DP}}(c)$ —i.e. that DAS, despite the alarming $c \rightarrow \infty$ result of Sutter et al. (2025), may be the more graceful failure at the finite capacities used in practice. We stress that this is a hypothesis to be tested, not a claim; the constraints above could equally be swamped by the sheer flexibility a non-linear ϕ buys, in which case DAS would fail *more sharply*. Either outcome is informative.

5 Experimental protocol

We instantiate the study on the **hierarchical equality task**, the toy setting of Sutter et al. (2025), for which an implementation already exists in this repository (a 3-layer MLP trained to decide whether two pairs of vectors are “both equal or both unequal”, together with linear-probe and linear-DAS pipelines and *randomised-label* controls). The task is ideal because the ground-truth intermediate variables (WX, YZ) are known, so “signal” is unambiguous.

Complexity axes. We add a complexity knob to each method. For *diagnostic probing*, we replace the single logistic-regression probe with a family of MLP probes of increasing depth and width ($c \in \{\text{linear}, 1\text{-hidden-layer}, 2\text{-hidden-layer}, \dots\}$), as in Pimentel et al. (2020). For *DAS*, we replace the orthogonal (rotation) alignment map with a family of RevNet maps of increasing depth L_{rn} and hidden size d_{rn} , as in Sutter et al. (2025). The data budget n is held *fixed* across the sweep so that we isolate the effect of c (in contrast to the existing sweeps, which vary n).

Measurements. At each complexity level and for several seeds we record (i) the *signal*: fit on correct targets (probe accuracy / IIA on the trained network); and (ii) the *spurious capacity* $\hat{\mathfrak{R}}_n(\mathcal{H}_c)$ of 1, estimated by re-fitting on randomised targets (random token labels for probing; a randomly relabelled / random target algorithm for DAS). We additionally repeat the DAS measurement on a *randomly initialised* network, which makes the spurious axis vivid: any IIA there is, by construction, spurious.

Analysis. We (1) plot signal and spurious capacity against c for each method; (2) plot the two methods together in the *signal-spurious plane* of section 3; and (3) fit a simple growth model $\hat{\mathfrak{R}}_n(c) \approx a c^b$ to read off, and compare, the empirical growth exponents b with confidence intervals. The deliverable is a statement of the form “per unit of spurious capacity, method X recovers more genuine signal than method Y on this task,” with the supporting curves.

What would falsify the framing. If signal and spurious capacity rise in lockstep for *both* methods (curves collapsing onto the diagonal of the signal-spurious plane), then capacity carries no method-distinguishing information and the unification, while tidy, would be practically empty. We regard making this falsifiable as a feature.

6 Relation to prior work, and a caveat on consistency

The control-task literature is the obvious antecedent. Hewitt and Liang (2019) introduce *selectivity*—task accuracy minus control-task accuracy—as a criterion for choosing probes, and Voita and Titov (2020) propose minimum description length as a complexity-aware alternative. Our spurious-fitting capacity is, at a single complexity level, exactly a control-task score; the novelty here is (i) reading it as a Rademacher complexity, (ii) lifting it from diagnostic to *causal* probing, and (iii) studying its *scaling* rather than its value at one point.

We must, however, confront a tension with Pimentel et al. (2020), which argues *against* selectivity as a probe-*selection* criterion: penalising a complex probe for its control-task accuracy artificially discards capacity that yields a tighter information estimate. We do not retract that argument; we repurpose the quantity. Selectivity is a poor *objective* to optimise—one should still pick the best-performing probe for estimation—but it is an excellent *descriptive diagnostic* of how much a method *could* fool itself, and its growth rate is a property of the method, not a knob to tune. Measuring fragility is not the same as selecting against it.

Finally, a caveat on what 1 does and does not capture. Rademacher complexity is a statement about *finite-sample* memorisation (failure mode (b) of subsection 3.2). It is a tractable *correlate* of, but not identical to, the *population-level* vacuity that the two theorems prove (failure mode (a)). A method could have vanishing spurious capacity at finite c and still be asymptotically vacuous. We therefore present capacity as a measurable proxy for one—arguably the more practically dangerous—axis of spuriousness, not as a complete account of it.

7 Conclusion

We have argued that the diagnostic critique of [Pimentel et al. \(2020\)](#) and the causal critique of [Sutter et al. \(2025\)](#) are two views of one phenomenon—an empirical-risk maximiser flattering itself in proportion to its capacity—and that this phenomenon has a single, measurable name: the Rademacher complexity of the method’s hypothesis class on the representations. Separating population vacuity (where the methods are equally doomed) from finite-sample optimism (where they need not be), we proposed to compare diagnostic probing and DAS in a signal–spurious plane, and identified invertibility, the frozen network, and multi-intervention consistency as concrete structural reasons DAS *might* fail more gracefully. The experiment is small, the infrastructure largely exists, and the outcome—whichever way it falls—tells us something we currently only guess at: not whether our interpretability tools can fool us, but how quickly.

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